Nature of Approval	Issuing Authority	Current Status	Our Remarks
			conversion in accordance with the provisions of Rajasthan Tenancy Act 1955 and Rajasthan Land Revenue Act 1956 and the rules made there under. Thus, Land NA Approval is not required for the project.
Power Telecommunication Coordination Committee (PTCC) Route Approval from Railway	North Western Railway	Obtained vide Letter Reference No. SG/158/NWR/PTCC/1026, dated 22-01-2025	As per the approval obtained, at present there is no Railway Track within the vicinity of 8 km from the proposed Power Line, hence, PTCC Approval for Development of 400 kv Transmission Line is provided for.
Power Telecommunication Coordination Committee (PTCC) Route Approval from Ministry of Defence	Director General of Signals (Signals 7) General Staff Branch Integrated Head Quarter of Ministry of Defence (Army), New Delhi	Obtained vide Letter Reference No. B/46937/Sign-7(b)/4365, dated 19-02-2025	NOC received from Ministry of Defence for Development of 400 kv Transmission Line.
Forest No Objection Certificate for Development of Transmission Line	Office of the Deputy Forest Conservator, Jaisalmer	Obtained vide Letter Reference No. क्रमांक एफ ( ) ऊबसं/तकनीकी/2024–25/, dated 18-03-2025	NOC obtained from Forest Department for Development of 400 kv Transmission Line.

Nature of Approval	Issuing Authority	Current Status	Our Remarks
Height Clearance for Laying of Transmission Line	Airport Authorities of India (AAI)	Obtained vide Letter Reference No. DEES/WEST/P/021125/1558984, dated 28-03-2025 and valid till 27-03-2033	NA
In-Principle Grant of Connectivity	Central Transmission Utility of India Limited (CTUIL)	Obtained vide Letter Reference Nos. CTU/N/Conn-INT-1A/0212100036 and CTU/N/Conn-INT-1A/2200000020, dated 23-11-2023 and 23-11-2023 respectively	Obtained for 600 MW
Final Grant of Connectivity	Central Transmission Utility of India Limited (CTUIL)	Obtained vide Letter Reference Nos. CTU/N/5/Conn-INT-1C/0212100036 and CTU/N/5/Conn-INT-1C/2200000020, dated 12-09-2024 and 12 respectively	Obtained for 600 MW
Temporary General Network Access (GNA) Approval	Central Electricity Regulatory Commission (CERC)	NA	As per Central Transmission Utility of India Limited, the Interim Date of Commissioning of Augmentation Infrastructure is 28-02-2026 and the final date is 22-08-2026. The company would be required to obtain Temporary General Network Access (GNA) Approval, prior to Project Commissioning in order to evacuate power before completion of Augmentation Infrastructure by Power Grid Corporation of India Limited.
First Time Charging Approval	Central Electricity Regulatory Commission (CERC)	NA	To be obtained prior to Commissioning

Nature of Approval	Issuing Authority	Current Status	Our Remarks
Project Commissioning Certificate	Rajasthan Renewable Energy Corporation Limited (RRECL)	NA	To be obtained prior to Commissioning

Our Comments:

- *Requisite Approvals required at this stage of the project is in possession of the Company.*
- *The present proposal is an Inter State Transmission System (ISTS) Project, wherein, Augmentation Infrastructure is being developed by Power Grid Corporation of India Limited. As per update by Central Transmission Utility of India Limited (CTUIL) vide its letter reference no. CTU/N/5/Conn-INT-1C/2200000020, dated 12-09-2024, the Augmentation Infrastructure is scheduled to be Commissioned by 22-08-2026. Further, as per Minutes of Meeting of the 34th Joint Co-ordination Committee Meeting for Generation & Transmission Projects of Northern Region issued by CTUIL vide Reference No. CTU/PMG/34th JCC-NR/MoM, dated 01-09-2025, the Augmentation Infrastructure is likely to be commissioned and operational by 22-08-2026. Thus, at present no delay is being anticipated for completion of the same.*
- *Balance Approvals would be obtained at a later stage.*

7 INDUSTRY HIGHLIGHTS

7.1 Introduction

Energy is primarily available in two forms vis-à-vis Renewable and Non-renewable sources. Off the forms available to us we still are dependent primarily on non-renewable sources.

Renewable energy derives its sources from Solar, Wind, Tidal Power and Hydro. The energy sources for renewable energy generation are available in abundance and shall be available for generations to come without the threat of there being depleted.

With about 300 clear and sunny days in a year, the calculated solar energy incidence on India's land area is about 5 quadrillion kilowatt-hours (kWh). The solar energy available in a single year exceeds the possible energy output of all of the fossil fuel energy reserves in India.

The demand for power around the world is steadily increasing with enhanced living standards, supported by rapid urbanization. There has been a spurt in the demand for renewable energy since conventional electricity generation methods such as thermal power plants are getting exhausted gradually. In India, solar and wind are the most popular renewable sources of energy. At present, India ranks third in the Renewable Energy Country Attractive Index. The renewable energy market is constantly expanding through efficient collaboration between the government and the private sector.

India has committed to a significant shift in its energy mix, aiming for 50% of its installed power capacity to come from non-fossil fuel sources by 2030. However, in 2023, fossil fuels still dominated India's electricity generation, accounting for 78% of the total. Coal was the primary source, contributing 75% of the generated electricity, making India the second-largest coal-dependent electricity producer in the G20 after South Africa. While clean energy sources like solar, wind, and hydro made up 22% of India's electricity generation mix, this is notably lower than the global average of 39% and the Asian average of 32%.

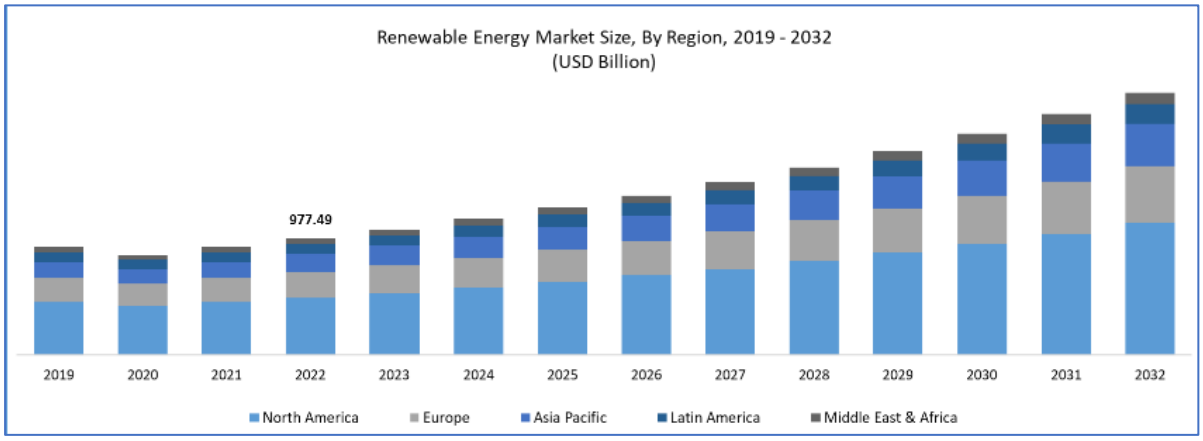
7.2 Renewable Energy & World

The global Renewable Energy market was valued at USD 1056.27 billion in 2023 and is expected to grow at a CAGR of 8.6% during the forecast period (2024-2032).

Global renewable capacity is expected to grow by **2.7 times by 2030**, surpassing countries' current ambitions by nearly 25%, but it still falls short of tripling. Climate and energy security policies in nearly 140 countries have played a crucial role in making renewables cost-competitive with fossil-fired power plants. This is unlocking new demand from the private sector and households, while industrial policies that encourage

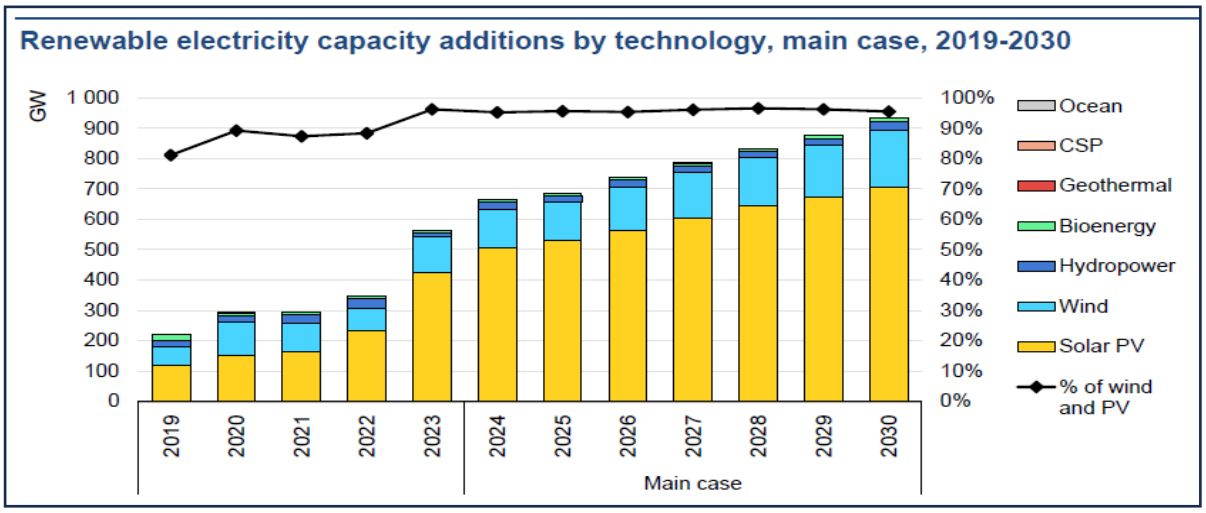
local manufacturing of solar panels and wind turbines are fostering domestic markets. However, this is not quite sufficient to reach the goal of tripling renewable energy capacity worldwide established by nearly 200 countries at the COP28 climate summit

The embrace of low-carbon fuels and the implementation of rigorous environmental regulations in many developed nations have significantly propelled the renewable energy sector. Over the recent years, the energy generation market has witnessed a rise in the installed capacity of renewable sources, spurred by growing environmental apprehensions and the imperative to alleviate the detrimental effects of Greenhouse Gases (GHG). This has played a crucial role in the growth of the solar and wind energy sectors.



Source: Polaris Market Research

Renewable capacity expansion in the next five years is anticipated to be much faster than what was expected just a year ago. The upward revision is mainly driven by China, the European Union, the United States and India, which are all implementing existing policies and regulatory and market reforms. The renewable capacity additions estimation in the next few years (as per IEA Report'2024) is as follows –

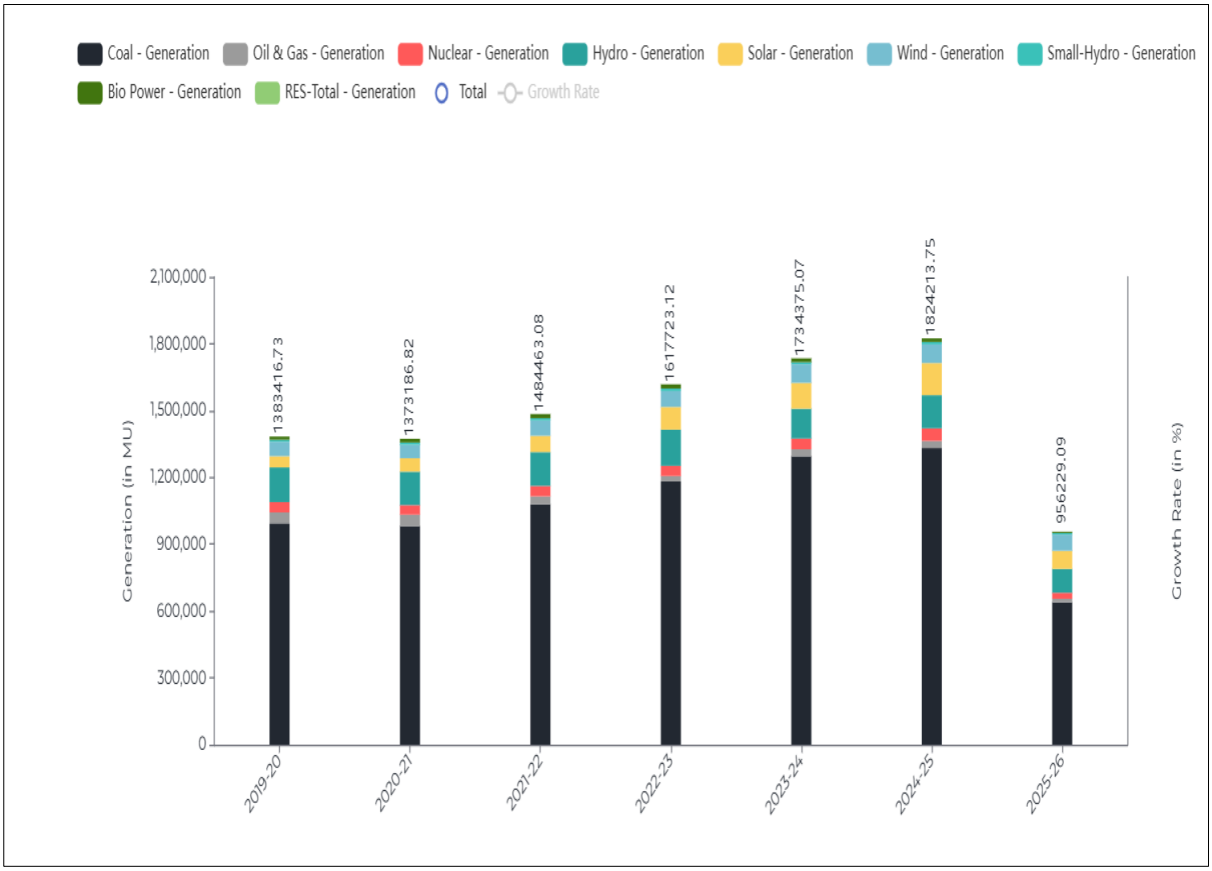


As per International Agency Renewable (IEA) Report of 2024, the share of renewable energy in power generation in the Asia Pacific is expected to increase from 22% in 2023 to 32% in 2030, which is in line with the global trend. As a result, the region retains its position as the third-smallest renewable energy market, surpassing only Eurasia and the Middle East and North Africa.

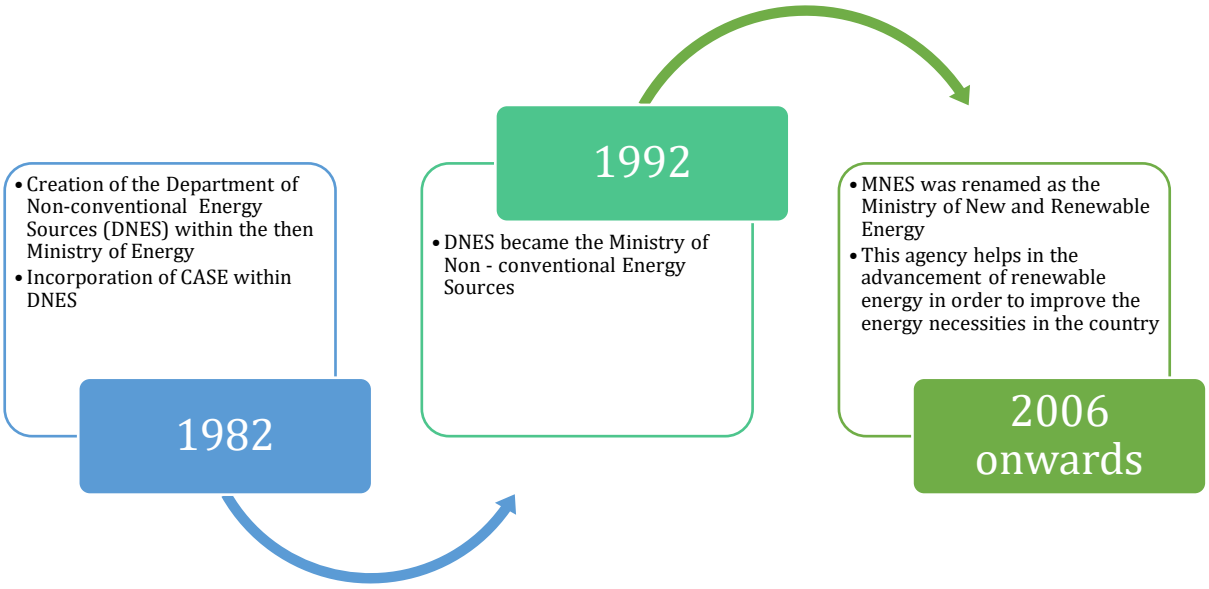
7.3 Renewable Energy in India

Growing population and increased energy demands of India, has made it imperative to look away from traditional power generation sources and increase reliance on renewable energy generation.

India's energy demand is primarily fulfilled by the Thermal Power Sector, with Coal Fired Power Plants generating nearly 638.88 Billion Units as on 30th September'2025. A share of source wise power generation till 30th September'2025 is provided below:



With a view to focus reduce reliance on thermal power generation and focus on renewable power generation, Ministry of New & Renewable Energy (MNRE) was established. Aim of MNRE is developing and deploying renewable energy to supplement energy requirements. A brief history of MNRE is provided as under:

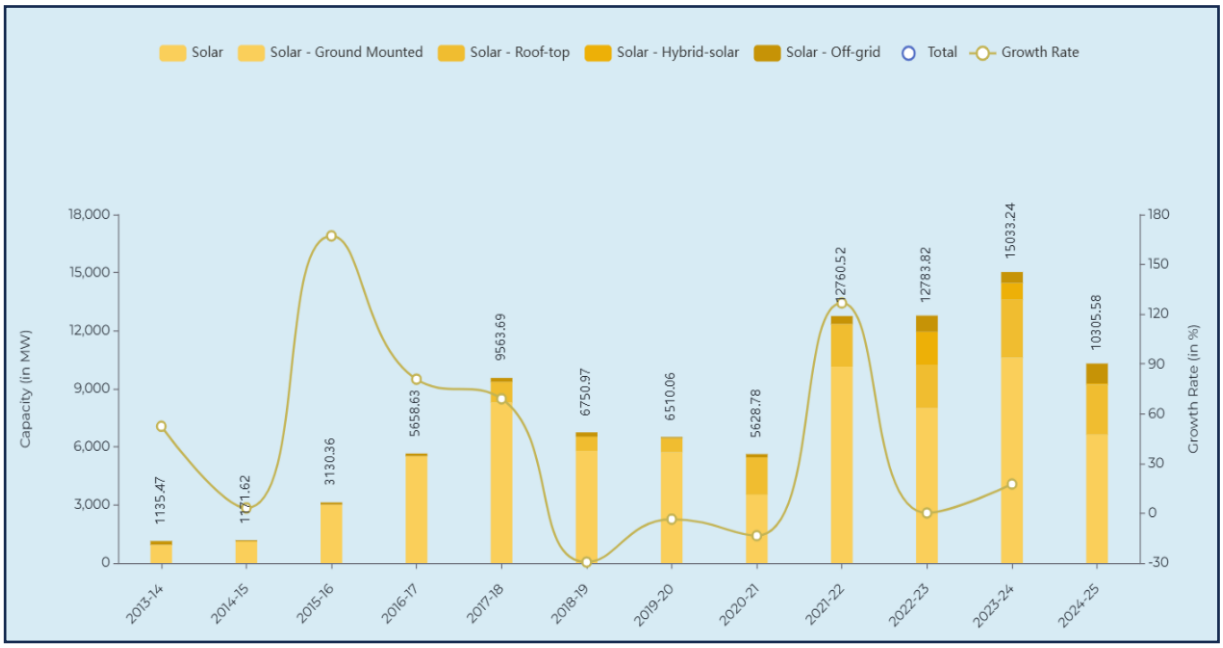


As per India Climate & Energy Dashboard functioning under the NITI Aayog, as on September'2025, the total installed capacity of power generation from renewable energy sources such as Solar, Wind, Hydro, and Small Hydro stands at 500889.23 MW.

7.4 Solar Energy in India

India is blessed with about 300 clear and sunny days in a year, the calculated solar energy incidence on India's land area is about 5 quadrillion kilowatt-hours (kWh). The solar energy available in a single year exceeds the possible energy output of all of the fossil fuel energy reserves in India. The total potential of India's Solar Energy Capacity is 7,48,990 MW, against which as on 31-10-2024 the installed capacity is only 92119.18 MW (being at just 12.30%).

During the last decade the trend for installed capacity of Solar Power Plants in India has shown an upward trend. From FY2013-14 to FY2023-24, the net capacity addition has grown at CAGR of 26.47%. A chart depicting the net capacity addition of Solar Power Plants in India from FY14 to FY25 (till 31-10-2024) is provided hereunder:



Source: https://iced.niti.gov.in/energy/electricity/generation/capacity

The National Institute of Energy estimated the country's solar potential at 748 GW, assuming solar PV modules cover 3% of the geographical surface. India is a perfect location for solar energy because of its location. It has 300 days of sunshine each year, with daily peak electricity use being in the evenings and a seasonal peak in the summer.

Table 42: State Wise Installed Capacity of Solar Power as on 31-10-2024

State/UT	Ground Mounted Solar (MW)	PM-Surya Ghar Yojna (Solar Rooftop) (MW)	Off-grid Solar/KUSUM (MW)	Solar Power Total (MW)
Andhra Pradesh	4337.67	224.88	88.34	4650.89
Arunachal Pradesh	0.00	0.00	6.77	1115.00
Assam	126.00	45.33	9.44	180.77
Bihar	146.06	90.00	21.28	257.34
Chhattisgarh	776.60	98.45	390.73	1265.78
Goa	45.50	0.00	1.41	47.86
Gujarat	10223.80	4340.05	89.09	15305.26
Haryana	266.80	722.03	0.00	916.36
Himachal Pradesh	80.68	22.03	34.58	137.29
Jammu & Kashmir	3360.00	0.00	67.00	3623.82
Jharkhand	21.00	93.04	67.00	181.04
Karnataka	8261.33	632.13	36.64	8930.10
Kerala	913.62	24.93	68.96	1261.76
Ladakh	89.00	0.00	0.00	141.29

State/UT	Ground Mounted Solar (MW)	PM-Surya Ghar Yojna (Solar Rooftop) (MW)	Off-grid Solar/KUSUM (MW)	Solar Power Total (MW)
Madhya Pradesh	3722.99	423.66	102.04	4248.69
Maharashtra	4747.43	2651.81	734.33	8133.57
Manipur	0.95	0.00	1.41	124.24
Meghalaya	0.00	0.21	4.07	395.11
Mizoram	1.96	0.00	6.39	135.82
Nagaland	1.00	0.00	2.17	110.84
Odisha	507.80	58.39	42.19	608.38
Punjab	886.27	408.16	81.36	1375.79
Rajasthan	20473.11	1294.57	805.45	24553.13
Sikkim	2282.00	0.00	137.29	2344.67
Tamil Nadu	8462.92	792.17	68.96	9324.05
Telangana	4360.49	472.90	8.71	4842.10
Tripura	4.78	0.00	11.08	36.94
Uttar Pradesh	2700.02	271.93	315.03	3286.98
Uttarakhand	273.71	19.96	19.96	592.07
West Bengal	230.20	67.13	13.14	310.47
Andaman & Nicobar Islands	35.16	0.27	1.41	35.16
Chandigarh	75.51	0.81	1.46	75.51
Dadra & Nagar Haveli and Daman & Diu	48.12	0.00	0.00	48.12
Delhi	277.09	9.84	1.46	288.39
Lakshadweep	0.00	0.00	0.00	2.52
Puducherry	0.18	0.00	0.00	52.27
Others	45.01	0.00	45.01	45.01
Total	**77784.76**	**12764.68**	**3284.16**	**98984.39**

Source: https://mnre.gov.in/en/physical-progress/

7.5 Stabilisation of Solar Module Prices

Despite domestic manufacturing capacity, India heavily relies on solar module imports, currently fulfilling 80-85% of its demand. This reliance stems from a limited domestic manufacturing base for critical components like polysilicon ingots and wafers, necessitating costly imports. Furthermore, the installed capacity for solar cells within India is insufficient, forcing module manufacturers to import cells primarily from China, followed by Hong Kong and Malaysia.

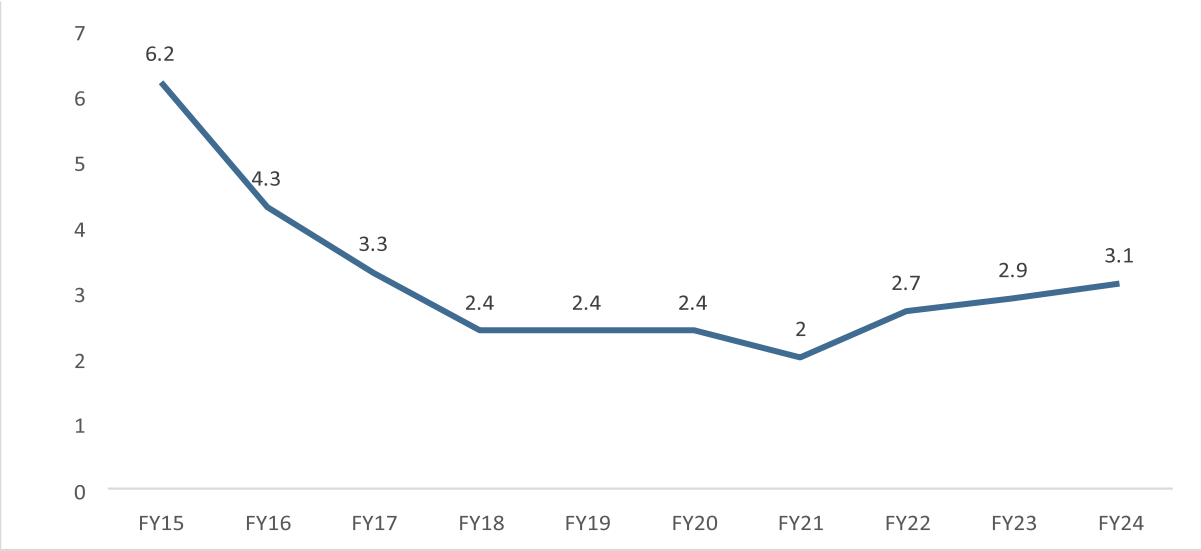
The government has implemented measures to bolster the domestic solar industry, including increased import duties. This, coupled with the commissioning of new polysilicon production capacities, has contributed to a decline in module prices in the first half of FY24 compared to the previous year. This price reduction is expected to accelerate solar installation rates.

While global module prices began to fall in 2023 following the easing of supply chain disruptions, they remain sensitive to supply from Chinese Original Equipment Manufacturers (OEMs) who dominate the solar PV manufacturing value chain.

However, global inventory levels and expansion plans by Indian players, driven by the Amended and Liberalized Manufacturing Policy (ALMM) and Basic Custom Duty, are expected to keep prices relatively subdued.

7.6 Trend of Solar Power Tariff

Solar tariffs in India have become highly competitive and achieved grid parity. This progress is attributed to advancements in technology, economies of scale, and a substantial reduction in the cost of solar cells and modules. A notable decline in solar tariffs has been observed, from INR 6.2 per kWh in Fiscal Year 2015 to INR 3.1 per kWh in Fiscal Year 2024.



Source: MNRE, CareEdge Research
Note: For FY22,FY23 and FY24, tariffs represent an average of projects bid during the resp. periods.

Exhibit: Trend in Solar Tariff

The bid tariff rates during FY23 were around Rs 2.7 - 3 per unit. Whereas in FY22, the bid tariff rates were around Rs. 2.7 per unit, which is 36% higher than FY21 primarily due to the rising equipment pricing, raw material costs, government duties, and interest rates.

Despite this, the bid tariff rates remained lower than that of FY15 levels. The current tariffs in the Indian solar sector (hovering at Rs2.50 -3/kWh) have stabilised at rates about 20-30% below the cost of existing thermal power in India, and up to half the price of new coal-fired power.

Table 43: Recent Bidding Trends

SN	Project Name	FY24	Tariff (Rs. /kWh)
1	NTPC Pan India Solar Tranche-II, 1,500MW	Mar-24	2.60
2	SECI Pan India Solar Tranche-XIV, 1,500 MW	Mar-24	2.58
3	MSAPL Maharashtra Solar, 5000 MW (Agricultural Feeder Scheme)	Mar-24	3.09
4	MSAPL Maharashtra Solar Tranche-I, 208MW (Agricultural Feeder Scheme).	Mar-24	3.30
5	MSAPL Maharashtra Solar Tranche-I, 196MW (Agricultural Feeder Scheme)	Mar-24	3.08
6	MSAPL Maharashtra Solar, 328MW (Agricultural Feeder Scheme)	Mar-24	3.09
7	MSAPL Maharashtra Solar, 126MW (Agricultural Feeder Scheme)	Mar-24	3.08
8	MSAPL Maharashtra Solar Tranche-I, 208 MW (Agricultural Feeder Scheme)	Mar-24	3.09
9	MSAPL Maharashtra Solar, 155MW (Agricultural Feeder Scheme)	Mar-24	3.09
10	MSAPL Maharashtra Solar Tranche-II, 196MW (Agricultural Feeder Scheme)	Mar-24	3.10
11	MSAPL Maharashtra Solar Tranche-II, 170MW (Agricultural Feeder Scheme)	Mar-24	3.10
12	MSAPL Maharashtra Solar Tranche-II, 234MW (Agricultural Feeder Scheme)	Mar-24	3.10
13	MSAPL Maharashtra Solar Tranche-II, 306MW (Agricultural Feeder Scheme)	Mar-24	3.10
14	MSAPL Maharashtra Solar Tranche-II, 170MW (Agricultural Feeder Scheme)	Mar-24	3.10
15	MSAPL Maharashtra Solar Tranche-II, 225MW (Agricultural Feeder Scheme)	Mar-24	3.10
16	SECI Pan India Solar Tranche-XIII, 1,500 MW	Mar-24	2.57
17	GUVNL Gujarat Solar Tranche-XXIII, 1,125MW	Mar-24	2.65
18	SJVN Pan India Solar, 1,500MW	Feb-24	2.53
19	GUVNL Gujarat Solar Tranche-XXI, 600MW	Jan-24	2.55
20	NTPC Pan India Solar Tranche-I, 1,500MW	Jan-24	2.63
21	GUVNL Pan India Solar Tranche-XXII, 500MW	Dec-23	2.64
22	PFCCL Pan India Solar, 1,250MW	Dec-23	2.54
23	SECI Pan India Solar Tranche-XIII, 1,000 MW	Dec-23	2.53
24	TANGEDCO Tamil Nadu Solar, 420MW (KUSUM A)	Nov-23	3.28

SN	Project Name	FY24	Tariff (Rs. /kWh)
25	NHPC Pan India Solar 3,000 MW	Nov-23	2.52
26	RVUNL Rajasthan Solar 810 MW	Oct-23	2.64
27	PEDA Multiple States Solar 18 MW	Sep-23	3.27
28	APDCL Assam Solar Tranche II 50 MW	Aug-23	3.92
29	APDCL Assam Solar 200 MW	Aug-23	3.90
30	APDCL Assam Solar 70 MW	Aug-23	3.92
31	RUVNL Pan India Solar Storage Hybrid 1,200 MW	Aug-23	6.68
32	RUMS Madhya Pradesh Floating Solar Tranche II 300 MW	Aug-23	3.84
33	PSPCL Punjab Solar 200 MW	Jul-23	2.75
34	PSPCL Pan India Solar Tranche I 1,000 MW	Jul-23	2.53
35	SECI Pan India Solar Tranche XI 2,000 MW	Jul-23	2.60
36	GUVNL Gujarat Solar Tranche XX 800 MW	Jul-23	2.70
37	JDWN Rajasthan Solar 27 MW	Jun-23	3.55
38	JDWN Rajasthan Solar 31 MW	Jun-23	3.48
39	AVVNL Rajasthan Solar 40 MW	May-23	3.44
40	BMC Madhya Pradesh Solar 21 MW	May-23	3.47
41	RUVNL Rajasthan Solar 1,000 MW	May-23	2.61
42	GUVNL Gujarat Solar Tranche XVII 600 MW	May-23	2.88
43	GUVNL Pan India Solar Tranche XIX 500 MW	Apr-23	2.72
44	MSEDL Maharashtra Solar Tranche X 500 MW	Apr-23	2.87
45	RECPDCL Pan India Solar 500 MW	Apr-23	2.69
46	RECPDCL Pan India Solar Tranche I 1,250 MW	Apr-23	2.55

Source: CareEdge Research Report

A comparison of Tariff for other Group Captive Projects is given below:

Table 44: Tariff for other Group Captive Projects

Particulars	Company A	Company B	Company C	Company D	Company E	Company F
Project Location	Ghatanji, Maharashtra	Latur, Maharashtra	Kalaburagi, Rajasthan	Mahidad, Gujarat	Barmer, Rajasthan	Tuticorin, Tamil Nadu
Project Capacity	26.40 MWdc	75 MWdc	185.90 MWdc	70 MWdc	75 MWdc	128 MWdc
Tariff	Rs.3.45 per unit	Rs.3.25 per unit	Rs.3.18 per unit	Rs.4.00 per unit	Rs.2.97 per unit	Rs.2.99 per unit

Source: LSI Internal Data Library

7.7 Challenges of Solar Power Plant

Although installation of Solar Power Plant is on the rise and is the need of the hour, there are some challenges too, some of the challenges are detailed as below:

Table 45: Challenges of Solar Power Plant

Challenges	Brief
Dependency on Weather Conditions	Supply of electricity relies on the weather, making it difficult for capacity estimation, thus making solar an unpredictable source of electricity
Land availability	One of the biggest challenges in the solar sector is the availability of land. Solar PV plants require a large amount of contiguous land to set up. This is particularly challenging in a country like India, given the land ownership issues, local politics and terrains
Solar Waste	Although life of solar plants is typically 25 years, the modules do get damaged, thus contributing to solar waste
Local Community Conflicts	There can be conflicts with local communities regarding use of land

7.8 Government Schemes and Subsidies for Renewables

As part of its Nationally Determined Contribution (NDC) to the Paris Agreement obligations, the government stated that by 2030, reduction of the emissions intensity of GDP by 45% below 2005 levels, and raise the percentage of non-fossil fuels in total capacity to 50% and increase the share of non-fossil power capacity to 50%. Hence the government has pushed towards renewable capacity additions through policy initiatives like JNNSM, obligations of RPO, setting up of SECI, etc.

Under Section 86(1) (e) of the Electricity Act 2003 and the National Tariff Policy 2006, Renewable Purchase Obligation (RPO), is a mechanism by which the obligated entities are obliged to purchase a certain percentage of electricity from renewable energy sources, as a percentage of the total consumption of electricity or buy, in lieu of that, renewable energy certificates (REC) from the market.

RPOs were earlier categorized as solar and non-solar RPOs. However, as per the latest targets, RPOs are categorized as Wind RPO, Hydro RPO, Distributed RPO and Others. Obligated entities (which include distribution companies (or DISCOMs), open access consumers and captive power producers) are obligated to purchase a minimum share of their electricity from renewable energy sources as per RPO targets. The RPO target set for FY18 was 14.25% which was gradually increased to 43.33% for FY30.

a) Green Energy Corridor

The Green Energy Corridor scheme was launched in 2015 for the setting up of transmission and evacuation infrastructure to facilitate the evacuation of electricity from renewable energy projects. The Intra state transmission system (ISTS) projects have been sanctioned to eight renewable energy states, i.e., Tamil Nadu, Rajasthan, Karnataka, Andhra Pradesh, Maharashtra, Gujarat, Himachal Pradesh, and Madhya Pradesh for evacuation of over 20,000 MW of renewable energy.

As of October, 2023, the project on Green Energy Corridor (GEC) Phase-II – Inter-State Transmission System (ISTS) for 13 GW Renewable Energy Project in Ladakh was approved. The project is targeted to be set up by FY 2029-30 with total estimated cost of Rs.20,773.70 crore and Central Financial Assistance (CFA) @ 40 percent of the project cost i.e. Rs.8,309.48 crore.

This project is in addition to Intra-State Transmission System Green Energy Corridor Phase-II (InSTS GEC- II), which is already under implementation in the States of Gujarat, Himachal Pradesh, Karnataka, Kerala, Rajasthan, Tamil Nadu and Uttar Pradesh for grid integration and power evacuation of approx. 20 GW of RE power and is expected to be completed by 2026. The InSTS GEC-II scheme targets addition of 10753 ckm of transmission lines and 27546 MVA capacity of substations having estimated project cost of Rs.12,031.33 crore and CFA @33%, i.e. Rs.3970.34 crore approved in December 2022.

b) Round-the-Clock-Power (RTC) for RE projects

The round-the-clock power mechanism is a bundling of power that has been bought by the government in order to overcome the issues of intermittency and low capacity utilization of transmission infrastructure. Here, the RE power is bundled with other sources and/or storage. Further, the Government is focusing towards promoting increased adoption of renewable energy and use of green hydrogen as envisaged in the National Green Hydrogen Mission, the Ministry of New and Renewable Energy is working on the modalities for promoting the use of green hydrogen in supporting round-the-clock electricity.

c) Competitive Bidding Guidelines for Solar and Wind Projects

The bidding guidelines have been issued for long-term procurement of power to promote competitive procurement from solar and wind and also to protect consumer interests. The guidelines for tariff-based competitive bidding process for procurement of power from grid-connected solar PV power projects were issued vide resolution 3rd August 2017 while the guidelines for tariff-based competitive bidding process for procurement of power from grid-connected wind power projects were issued vide resolution dated 8th Dec 2017.

d) Waiver of ISTS Charges

Ministry of Power has issued an order for an extension to the inter-state transmission system (ISTS) charges waiver on solar and wind energy projects commissioned up to 30 June 2025. Waiver of ISTS charges shall also be allowed for hydro-pumped storage plant and battery energy storage system projects to be commissioned up to 30th June 2025 following some conditions.

ISTS waiver would be allowed for trading electricity generated and supplied from solar, wind, pumped hydro, and Battery Energy Storage Systems (BESS) in the green term ahead market (GTAM) till 30th June 2023 and the arrangement would be reviewed on an annual basis depending on future development in the power market.

As per the notification issued by the Ministry of Power, a complete waiver of ISTS charges has been given for offshore wind power projects commissioned on or before 31st December, 2032 for a period of 25 years from the date of commissioning of the Project.

e) Must Run Status

In line with the Electricity Act 2003 and the Electricity Grid Code 2010, wind and solar power have the 'must-run status'. The term 'must run status' refers to the notion that electricity evacuation from solar and wind power facilities should not be limited for reasons other than grid safety, equipment or people safety, merit order dispatch, or other commercial concerns.

7.8.1 Government Schemes and Subsidies for Renewables

For Solar Projects

Jawaharlal Nehru National Solar Mission	- Jawaharlal Nehru National Solar Mission (JNNSM) is one of the primary missions under India's National Action Plan on Climate Change. JNNSM is a major initiative by the Indian government to encourage environmentally sustainable growth while addressing India's energy security issues. To meet this goal, the Indian Government has implemented a number of policies, including the Solar Park Scheme, Pradhan Mantri Kisan Urja Suraksha evam Utthan Mahabhian (PM KUSUM), Central Public Sector Undertaking (CPSU), Grid Connected Solar Rooftop Schemes, Domestic modules production, REC, RPO, waiver of ISTS charges , etc.
International Solar Alliance	- The International Solar Alliance (ISA) is a treaty-based inter-governmental organization working to create a global market system to tap the benefits of solar power and create clean energy applications. The aim of ISA is to pave the way for future solar generation, storage and technologies for the member countries by mobilizing over USD 1000 billion by 2030. The achievement of ISA's objective will help the

	member countries fulfil the Nationally Determined Contributions (NSC) commitments.
Pradhan Mantri Kisan Urja Suraksha Evam Utthan Mahabhiyan (PM KUSUM)	- The PM-KUSUM programme is to supply renewable energy to over 3.5 million farmers by solarizing their agriculture pumps. The PM-KUSUM programme intends to build grid-connected ground-mounted solar power plants (up to 2 MW) totalling 10 GW under Component A, and 2 million freestanding solar pumps under Component B, and solarize 1.5 million grid-connected agricultural pumps under Component C. All components combined would support installation of additional solar capacity of 30.80 GW. - As of January 2024, 165.28 MW capacity solar power plants were installed under the scheme's Component- A, about 2,95,823 stand-alone solar pumps were installed under Component-B and 7,384 pumps were reported solarised under individual pump solarisation variant of Component-C.
Roof Top Solar (RTS) Programme	- Rooftop solar power (RTS) is a rooftop solar system that generates electricity for Kenyan households and public buildings. Rooftop Phase I of this initiative began on December 30, 2015, with incentives and subsidies offered for the residential, institutional, and social sectors. Achievement-based incentives were also offered for the government sector. Rooftop Phase II began in February 2019 with the goal of reaching a total capacity of 40,000 MW by 2022. RTS has built approximately 3.7 GW of capacity so far, with another - GW under construction in the residential market. Central Financial Assistance is given at 40% for RTS systems up to 3 kW capacity and 20% for systems with capacities more than 3 kW. - Phase II of the Rooftop Solar Programme timelines have been extended up to 31.03.2026. As on November 2023, capacity installed undewr Phase-II stands at 2.65 GW.
PM-Surya Ghar: Muft Bijli Yojana	- The Government launched the scheme PM-Surya Ghar: Muft Bijli Yojana in February, 2024 with a total outlay of Rs. 750.21 Bn. This scheme aims to install rooftop solar and provide up to 300 units of free electricity every month for one crore households. - The scheme includes Central Financial Assistance (CFA) for Residential Rooftop Solar.
Solar Parks	- The Ministry of Power has introduced the Solar Parks programme with the objective of facilitating solar project

	developers to set up projects in a plug-and-play model. The scheme for the development of solar parks has a target capacity of 40 GW and all States and Union Territories are eligible for getting benefits under the scheme. - As of December 2023, 51 Solar Parks with an aggregate capacity of 37,740 MW have been sanctioned in 12 States in the country since launch of the Scheme i.e. December 2014. An aggregate capacity of 10,504 MW of solar projects have been commissioned in 20 Solar Parks till date.
Solar Cities	- Under this scheme, at least one city in each state of India is being developed as a solar city. Here, all the electricity needs of the city will be met through RE sources primarily from solar energy and all houses will have roof-top solar energy plants along with solar street lights and waste-to-energy plants. - The aim of the programme is to enable and empower urban local government to address the energy challenges at the city level and provide a framework and support to prepare a master plan including an assessment of the current energy situation, future demand, and action plans. - There is no separate fund allocation for solar city programme and the funds available under the different existing schemes can be utilized for development of solar cities. So far, 27 States/UTs have identified solar cities in their respective States/UTs.
Public Sector Undertaking (CPSU) Scheme	- The Cabinet Committee on Economic Affairs ("CCEA") has approved the MNRE's proposal for the implementation of the CPSU Scheme Phase-II for setting up 12,000 MW grid-connected solar PV power projects with VGF support of Rs. 858 million for self-use or use by government or government entities, of both central and state governments. The scheme mandates the use of both solar PV cells and modules manufactured domestically as per specifications and testing requirements fixed by the MNRE. - Under this scheme, around 8.2 GW of projects have been awarded, as of 31.12.2022. Out of which, around - 1.5 GW has been commissioned as of 31.12.2022 and the remaining is under implementation.

7.9 Renewable Energy Growth Drivers

Table 46: Renewable energy growth drivers

Government commitments	- Prime Minister Mr. Narendra Modi initially set the target of installing 175 GW of renewable energy capacity by 2030, but has now increased it to 450 GW. - US$ 2.4 billion National Hydrogen Mission for production of 5 MMT by 2030 and US$ 36 million additional in budget. - The Indian government's commitment to reaching net-zero emissions by 2070 and increasing its renewable energy target to 500 GW by 2030 at the COP26 summit has provided great support to the industry and spurred unprecedented growth. - In October 2021, the Ministry of Power announced a new set of rules aimed at reducing financial stress for stakeholders and safeguarding timely cost recovery in electricity generation. - In November 2021, the government announced future plans to increase the funding under the PLI scheme for domestic solar cells and module manufacturing to Rs. 24,000 crore (US$ 3.17 billion) from the existing Rs. 4,500 crore (US$ 594.68 million) to make India an exporting nation. - Solar city per state-approved and approved setting up 59 solar parks of 40 GW across the nation. The government is also giving a push to Floating PV Projects.
Investments	- Around US$ 2.8 trillion will be invested in energy in 2023 globally. More than US$ 1.7 trillion is going to clean energy, including renewable power, nuclear, grids, storage, low-emission fuels, efficiency improvements and end-use renewables and electrification. - In August 2022, Norfund, who manage the Norwegian Climate Investment Fund, and KLP, Norway's biggest pension company, signed an agreement to buy a 49% share of a 420 MW solar power plant in Rajasthan for Rs. 2.8 billion (US$ 35.05 million). - Investment in the renewable energy in India reached a record US$ 14.5 billion in FY22, an increase of 125% over FY21. - In 2022, India's renewable energy sector is expected to boom with a likely investment of US$ 15 billion, as the government focuses on electric vehicles, green hydrogen, and the manufacturing of solar equipment.
Favourable policies and incentives	- In April 2021, the Ministry of Power (MoP) released a draft of the National Electricity Policy (NEP) 2021 and has invited suggestions from all stakeholders such as Central Public Sector Undertakings, Solar Energy Corporation of India, power transmission companies,

	financial institutions such as the Reserve Bank of India, Indian Renewable Energy Development Agency, HDFC Bank, ICICI Bank, industrial, solar and wind associations and state governments. - Proposed solar cities and parks: Solar city per state-approved and approved setting up 59 solar parks of 40 GW across the nation. The government is also giving a push to Floating PV Projects. - National Green Hydrogen Mission: The Union Cabinet approved the National Green Hydrogen Mission with a total initial outlay of Rs. 19,744 crore (US$ 2.37 billion), including an outlay of Rs. 17,490 crore (US$ 2.10 billion) for the SIGHT programme, Rs. 1,466 crore (US$ 176.2 million) for pilot projects, Rs. 400 crore (US$ 48.1 million) for R&D, and Rs. 388 crore (US$ 46.6 million) towards other Mission components. - Atmanirbhar Bharat: PLI scheme in Solar PV manufacturing with financial outlays of Rs. 24,000 crore (US$ 2.88 billion) introduced under Atmanirbhar Bharat. Imposition of Basic Customs Duty of 25% on Solar Cell & 40% on Solar PV Modules w.e.f. 01.04.2022. - Off-shore Wind Energy: The medium and long-term target for off-shore wind power capacity additions, are 5 GW by 2022 and 30 GW by 2030. - Wind-Solar Hybrid Policy: In 2018, a national policy was announced to promote a large grid-connected wind-solar PV hybrid system for efficient utilization of transmission infrastructure and land. A way to address the intermittency challenge of one source of renewable power is to combine solar and wind, achieving better grid stability. It provides flexibility in the share of wind and solar components in hybrid projects, however, the capacity of one resource must be at least 25% of the rated power capacity of another resource

7.10 Road Ahead

India has set a target to reduce the carbon intensity of the nation's economy by less than 45% by the end of the decade, achieve 50% cumulative electric power installed by 2030 from renewables, and achieve net-zero carbon emissions by 2070. Low-carbon technologies could create a market worth up to $80 billion in India by 2030.

India's ambitious renewable energy goals are transforming its power sector. The rising population and widespread electrification in rural homes are fueling the demand for energy to power homes, businesses and communities. Clean energy will reduce pollution levels as villages become self-sustainable with their use of clean energy. In 2022, India's renewable energy sector is expected to boom with a likely investment of US$ 15 billion

this year, as the government focuses on electric vehicles, green hydrogen, and the manufacturing of solar equipment.

It is expected that by 2040, around 49% of the total electricity will be generated by renewable energy as more efficient batteries will be used to store electricity, which will further cut the solar energy cost by 66% as compared to the current cost. The use of renewables in place of coal will save India Rs. 54,000 crore (US$ 8.43 billion) annually. Around 15,000 MW of wind-solar hybrid capacity is expected to be added between 2020-25.

As per the Central Electricity Authority (CEA) estimates, by 2029-30, the share of renewable energy generation would increase from 18% to 44%, while that of thermal is expected to reduce from 78% to 52%. The CEA also estimates India's power requirement to grow to reach 817 GW by 2030.

8 REVIEW OF THE COST OF THE PROJECT

8.1 Review of the Cost of the Project

Cost estimation provides a vital input to the economic and financial evaluation of the Project. LSI has reviewed the cost estimates as provided by the company. The Project is estimated to be set up at a cost of ***Rs.3846.51 crore***. Components of the same is detailed as under:

Table 47: Components of Project Cost

Particulars	Rs. Crore
Land Cost	70.00
EPC Cost	2989.08
Shared Infra and Evacuation Sharing Cost	152.15
Total Hard Cost	**3211.23**
Contingencies	64.22
Preliminary & Pre-operative Expenses	109.28
Finance Cost	94.04
Interest During Construction	275.63
DSRA	80.32
Total Soft Cost	**623.49**
Total CAPEX	**3834.72**
Working Capital Margin	11.79
Total Project Cost	**3846.51**

8.2 Cost of Land

Cost of Land for the proposed project is envisaged at Rs.70.00 crore. Land acquisition is in a mix of leasehold basis (both Private and Government) along with outright purchase.

Our Comments:

- *Initially the cost of land was envisaged at Rs.41.40 crore which has now increased to Rs.70.00 crore. The increase in cost of land is primarily due to the following:*
  - *The project involves acquisition of Government Land of ~1142 acres. A District Level Committee Charges (DLC) is levied by the state Government. Over the period of project implementation, the DLC charges have increased by Rs.22000 per acre, thereby increase land cost.*
  - *Due to Change in Law, GST @18% on lease rentals being paid to the Lessors is being incurred on Reverse Charge Mechanism basis, thus, increase of land cost.*
  - *The project was initially envisaged to be set up using leasehold land, however, due to unavailability, the Company had to procure ~125 acres of land on outright purchase basis.*
  - *Increased upfront lease rentals from 2 years to 4 to 5 years.*

- *Out of the envisaged Cost of Land of Rs.70.00 crore, an amount of Rs.57.57 crore (as per Chartered Accountant Certificate dated 06-11-2025) has already been incurred towards the project.*

8.3 Cost of EPC

The company has entered into an EPC Contract with Serentica Renewables India Private Limited for execution of the project at a cost of ***Rs.2989.08 crore*** (details of EPC Contract is mentioned in Section 5.1.1 of this Report). The cost of EPC has been worked out as under:

Table 48: Components of EPC Cost

Particulars	Rs. Crore
Supply of Solar Power Generating System	2092.36
Services/ Works	896.72
Total EPC Cost	**2989.08**

8.4 Shared Infra and Evacuation Sharing Cost

Cost of Shared Infra and Evacuation Sharing is estimated at ***Rs.152.15 crore***. The company for implementation of the project has executed an Shaed Infra and Evacuation Sharing Agreement with Serentica Renewables India Private Limited (details of Evacuation Sharing Agreement is mentioned in Section 5.1.2 of this Report).

8.5 Contingencies

Contingency Provisions estimated for the project is Rs.64.22 crore. Contingency has been calculated @2% of Total Hard Cost of the Project of Rs.3211.23 crore.

8.6 Preliminary & Pre-operative Expenses

Preliminary & Pre-operative Expenses envisaged for the project is Rs.109.28 crore. This constitutes the following:

- TEV/ LIE/ LLC/ LIA Charges.
- Salary During Construction Period.
- Insurance During Construction Period.
- Approval Costs.
- Hedging Costs

8.7 Finance Cost

Finance Cost envisaged for the project is Rs.94.04 crore. Finance Cost includes cost towards Loan Processing Fees and Pre-payment Charges to be paid to existing lenders.

Our Comments:

- ***Prepayment Charges:*** *As per existing Sanction of REC Limited dated 16-01-2024, a prepayment charges of 2.00% is applicable in case the prepayment is made within 3 years from the date of 1st Disbursement. As per information provided by the Company, the current outstanding of REC is Rs.1830.31 crore, thus, a prepayment charges of ~Rs.44.00 crore (including GST) is envisaged for the project.*
- ***Upfront Fees:*** *As per existing Sanction of REC Limited, a transaction fee of 0.50% of the sanctioned term loan is applicable. Considering term loan amount of Rs.2971.27 crore, upfront fees of ~Rs.18.00 crore (including GST) is envisaged for the project.*
- *Balance Finance Cost of ~Rs.32.00 crore is towards other finance cost.*

8.8 Interest During Construction (IDC)

IDC for the project is estimated to be at Rs.275.63 crore. Components of IDC is given below:

Table 49: IDC

Drawl Period	Amount of Drawl	Amount of IDC
Qtr.2 FY2025	204.62	2.41
Qtr.3 FY2025	330.09	8.71
Qtr.4 FY2025	301.46	15.79
Qtr.1 FY2026	740.47	28.11
Qtr.2 FY2026	489.44	42.91
Qtr.3 FY2026	791.10	58.00
Qtr.4 FY2026	132.44	58.41
Qtr.1 FY2027	87.59	61.29
Total	**3077.21**	**275.63**

8.9 DSRA

DSRA equivalent to 1 quarter principal repayment and interest (i.e. Rs.80.32 crores) has been considered prior to COD of the project.